Chapter 0 — The Machinery: Representables, Yoneda, Day Convolution, and Kan Extensions

MacBeth

Abstract

This chapter assembles the pre-container tools the rest of the book leans on: representable functors and the Yoneda lemma, density of representables, left Kan extensions and the coend calculus, Day convolution, and the interaction of the Yoneda embedding with monoidal and closed structure. *Everything here is standard*, collected for self-containedness and to fix notation. We claim no originality: the Yoneda lemma is Yoneda's and Mac Lane's; Kan extensions and the coend calculus follow Mac Lane [3] Ch. X, Riehl [4], and Loregian [2]; Day convolution is Day's [1]. Our only contribution is arrangement — pedagogical order, and proofs written in our own words — and the deliberate foregrounding of two observations of Neil Ghani that the book's monoidal chapter will invoke as black boxes: that free coproduct completion *is* a left Kan extension along the representable embedding (§2), and that closed structure on a Day-convolution category is *forced on representables and extended by density* (§4). We are careful about the second: density determines the internal hom, but it does *not* follow that an arbitrary cocontinuous strong-monoidal functor preserves it (we give a one-line counterexample, Remark 4.3); preservation is a genuine, separate fact, true for the container-extension functor and checked directly in the monoidal chapter.

Contents

Conventions. **Set** is the category of sets and functions. For a locally small category $\mathcal{C}$ we write $\mathcal{C}(a, b)$ for its hom-set. All functors are covariant unless a superscript $^{\mathrm{op}}$ says otherwise. A *presheaf* on $\mathcal{C}$ is a functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$; a *copresheaf* is a functor $\mathcal{C} \to \mathbf{Set}$. We use $\cong$ for canonical (natural) isomorphism throughout, and we are careful to say *which* isomorphism whenever it matters.

1 Representable functors and the Yoneda lemma

Every construction in this book eventually reduces to a hom-set. The representable functors are the functors that *are* a hom-set, one argument held fixed, and they turn out to generate everything else. We start there.

Definition 1.1 (Representable copresheaf)**.** Fix a set A. The *covariant representable* at A is the functor

$$y^A \;:=\; \mathbf{Set}(A,-) \;:\; \mathbf{Set} \longrightarrow \mathbf{Set}, \qquad y^A(X) = \mathbf{Set}(A,X) = X^A,$$

sending a function $f\colon X \to Y$ to post-composition $y^A(f) = f \circ (-)\colon X^A \to Y^A$. A copresheaf $F\colon \mathbf{Set} \to \mathbf{Set}$ is *representable* if $F \cong y^A$ for some A; we call A the *representing object*.

Remark 1.2 (The container reading)**.** This book is about *containers*. A container is a pair (S,P) of a set S of *shapes* and a family $P\colon S \to \mathbf{Set}$ of *position sets*, and its *extension* is the functor

$$[\![S,P]\!] \;:\; \mathbf{Set} \to \mathbf{Set}, \qquad [\![S,P]\!](X) \;=\; \sum_{s\in S} \mathbf{Set}\big(P(s),X\big).$$

The *representable container* is the one with a single shape and A positions, $(S,P) = (1,\,A)$ (the family $1 \to \mathbf{Set}$ picking out A). Its extension is

$$[\![1,A]\!](X) \;=\; \sum_{s\in 1} \mathbf{Set}(A,X) \;=\; \mathbf{Set}(A,X) \;=\; X^A \;=\; y^A(X).$$

So representable copresheaves are exactly the extensions of single-shape containers, and the coproduct in the general formula is a coproduct *of representables* indexed by shapes. This is not a coincidence to be admired and forgotten: §2 and §3 show that "a container is a coproduct of representables" is the whole story, phrased first as a Kan extension and then as a Day convolution.

1.1 The Yoneda embedding and the Yoneda lemma

Definition 1.1 is uniform in A: a function $a\colon A' \to A$ induces, by *pre*-composition, a natural transformation $y^A \Rightarrow y^{A'}$. Thus $A \mapsto y^A$ is a functor $\mathbf{Set}^{\mathrm{op}} \to [\mathbf{Set},\mathbf{Set}]$, the *Yoneda embedding* (covariant version). We write it $y^{(-)}$. It is genuinely an embedding: fully faithful, as the Yoneda lemma will show.

Theorem 1.3 (Yoneda lemma)**.** *Let $F\colon \mathbf{Set} \to \mathbf{Set}$ be any functor and A any set. There is a bijection*

$$\Theta_{F,A} \;:\; \mathrm{Nat}(y^A,\,F) \;\xrightarrow{\;\cong\;}\; F(A),$$

natural in both F and A, given by evaluating a natural transformation at the identity: $\Theta(\alpha) = \alpha_A(\mathrm{id}_A)$.

Proof. We build the inverse and check both round trips.

Inverse. Given an element $x \in F(A)$, define a family of maps $\Psi(x)_X\colon y^A(X) = \mathbf{Set}(A,X) \to F(X)$ by

$$\Psi(x)_X(g) \;=\; F(g)(x), \qquad g\colon A \to X.$$

This is natural in X: for $h\colon X \to Y$ we must show $F(h) \circ \Psi(x)_X = \Psi(x)_Y \circ y^A(h)$. Applied to $g\colon A \to X$ the left side is $F(h)(F(g)(x)) = F(h \circ g)(x)$ by functoriality, and the right side is $\Psi(x)_Y(h \circ g) = F(h \circ g)(x)$. They agree, so $\Psi(x)$ is a natural transformation $y^A \Rightarrow F$.

Θ *after* Ψ. For $x \in F(A)$,

$$\Theta(\Psi(x)) = \Psi(x)_A(\mathrm{id}_A) = F(\mathrm{id}_A)(x) = \mathrm{id}_{F(A)}(x) = x.$$

Ψ *after* Θ. Let $\alpha\colon y^A \Rightarrow F$ and put $x = \Theta(\alpha) = \alpha_A(\mathrm{id}_A)$. We must show $\Psi(x) = \alpha$, i.e. for every X and every $g\colon A \to X$ that $F(g)(x) = \alpha_X(g)$. Naturality of α at the map $g\colon A \to X$ gives a commuting square

$$\begin{tikzcd}
\mathbf{Set}(A,A) \arrow[r, "\alpha_A"] \arrow[d, "g\circ(-)"'] & F(A) \arrow[d, "F(g)"] \\
\mathbf{Set}(A,X) \arrow[r, "\alpha_X"'] & F(X).
\end{tikzcd}$$

Chase id_A around it. Down-then-right gives $\alpha_X(g\circ\mathrm{id}_A) = \alpha_X(g)$; right-then-down gives $F(g)(\alpha_A(\mathrm{id}_A)) = F(g)(x)$. Hence $\alpha_X(g) = F(g)(x) = \Psi(x)_X(g)$, as required.

Thus Θ and Ψ are mutually inverse bijections. Naturality of Θ in F (along natural transformations $F \Rightarrow F'$) and in A (along functions $A' \to A$, which act on y^A by pre-composition) is a routine diagram chase of the same shape; we omit it. □

Corollary 1.4 (Full faithfulness)**.** *Taking* $F = y^B$ *in Theorem 1.3 gives* $\mathrm{Nat}(y^A, y^B) \cong y^B(A) = \mathbf{Set}(B, A)$*, natural in* A *and* B*. Hence the Yoneda embedding* $y^{(-)}\colon \mathbf{Set}^{\mathrm{op}} \to [\mathbf{Set}, \mathbf{Set}]$ *is fully faithful: natural transformations between representables are exactly (pre-composition by) functions, in the opposite direction.*

Example 1.5 (Yoneda in **Set**, concretely)**.** Take $A = 1$, the one-element set. Then $y^A = \mathbf{Set}(1, -)$ is naturally isomorphic to the identity functor, since $\mathbf{Set}(1, X) \cong X$. The Yoneda lemma reads $\mathrm{Nat}(\mathrm{Id}, F) \cong F(1)$: a natural self-map of the identity is pinned down by a single element of $F(1)$. For $F = \mathrm{Id}$ this recovers $\mathrm{Nat}(\mathrm{Id}, \mathrm{Id}) \cong 1$: the only natural transformation $\mathrm{Id} \Rightarrow \mathrm{Id}$ is the identity. Try to build a "doubling" natural transformation $X \to X$ and you cannot even start: there is no natural way to duplicate an anonymous element, precisely because id_1 is the only probe you are allowed and it must go to itself.

1.2 Density of representables

The Yoneda lemma says representables see everything. Density says representables *are* everything, once you are allowed colimits.

Theorem 1.6 (Density / co-Yoneda)**.** *Every functor* $F\colon \mathbf{Set} \to \mathbf{Set}$ *is canonically a colimit of representables. Concretely,* F *is the coend*

$$F \;\cong\; \int^{A \in \mathbf{Set}} F(A) \cdot y^A,$$

where $F(A) \cdot y^A$ *denotes the* $F(A)$*-fold copower (coproduct) of the representable* y^A*. Equivalently,* F *is the colimit of the canonical diagram*

$$\big(\, y^A \xrightarrow{\;\Psi(x)\;} F \,\big)_{(A,x),\ x \in F(A)}$$

indexed by the category of elements of F*; the coend presentation packages this diagram and its identifications.*

Proof sketch. This is the "density formula," dual to Yoneda; see Riehl [4, §6.2] or Loregian [2, Ch. 2]. The universal property is exactly Theorem 1.3 read backwards: a cocone from the diagram $F(A)\cdot y^A$ to a functor G is, by the copower and Yoneda adjunctions, a family of maps $F(A)\to \mathrm{Nat}(y^A, G)\cong G(A)$ natural in A, i.e. exactly a natural transformation $F\Rightarrow G$. Since cocones out of the coend biject naturally with maps $F\Rightarrow G$, the coend *is* F. The coend's coequalizer presentation

$$\coprod_{f\colon A\to A'} F(A)\cdot y^{A'} \;\rightrightarrows\; \coprod_{A} F(A)\cdot y^{A} \;\twoheadrightarrow\; F$$

records the identification "push x along f in F" = "reindex the representable along f," which is where naturality is spent. □

Remark 1.7. Density is the load-bearing fact for §2 and §4: the representables form a *dense generator* of $[\mathbf{Set},\mathbf{Set}]$. Any two of our constructions that agree on representables and both preserve the relevant colimits agree everywhere. We will use this uniformity of argument for Neil's observation (ii) — "the internal hom is determined on representables and extended by density."

2 Left Kan extensions

A Kan extension is the best approximation to "extend a functor along another functor." We need the left version and its pointwise coend formula, and then one theorem: for our purposes the free coproduct completion of a category *is* a left Kan extension along the representable embedding. This is the lemma the monoidal chapter cites for its Theorem C.

Definition 2.1 (Left Kan extension)**.** Let $K\colon \mathcal{C}\to\mathcal{D}$ and $F\colon\mathcal{C}\to\mathcal{E}$ be functors. A *left Kan extension* of F along K is a functor $\mathrm{Lan}_K F\colon \mathcal{D}\to\mathcal{E}$ together with a natural transformation $\eta\colon F\Rightarrow (\mathrm{Lan}_K F)\circ K$ that is universal from F: for every $G\colon\mathcal{D}\to\mathcal{E}$ and every $\gamma\colon F\Rightarrow G\circ K$ there is a *unique* $\bar\gamma\colon \mathrm{Lan}_K F\Rightarrow G$ with $(\bar\gamma K)\circ\eta=\gamma$.

$$\begin{tikzcd}
\mathcal{C} \arrow[r, "F"] \arrow[d, "K"'] & \mathcal{E} \\
\mathcal{D} \arrow[ur, "\mathrm{Lan}_K F"'] \arrow[u, phantom, ""] &
\arrow[from=1-1, to=1-2, phantom, shift right=3, "\eta\,\Downarrow"]
\end{tikzcd}$$

Equivalently, $\mathrm{Lan}_K(-)$ is left adjoint to precomposition $(-)\circ K\colon [\mathcal{D},\mathcal{E}]\to[\mathcal{C},\mathcal{E}]$.

Theorem 2.2 (Pointwise coend formula)**.** *If $\mathcal{C}$ is small, $\mathcal{E}$ is cocomplete, and $\mathcal{D}$ is locally small, then $\mathrm{Lan}_K F$ exists and is computed pointwise by the coend*

$$(\mathrm{Lan}_K F)(d) \;\cong\; \int^{c\in\mathcal{C}} \mathcal{D}(Kc,\, d)\cdot F(c),$$

where $\mathcal{D}(Kc,d)\cdot F(c)$ is the $\mathcal{D}(Kc,d)$-fold copower of the object $F(c)\in\mathcal{E}$.

Reference. Standard; Mac Lane [3, Ch. X, §4], Riehl [4, §6.2], Loregian [2, Ch. 2]. The universal property of Definition 2.1 follows because maps out of the copower/coend transpose, via the co-Yoneda density argument, to natural transformations, exactly as in the proof of Theorem 1.6. □

2.1 Free coproduct completion *is* a left Kan extension

We now single out the instance the book uses. Recall the container category.

Notation 2.3 (**Cont** and $\mathbf{Fam}(\mathbf{Set}^{\mathrm{op}})$)**.** A morphism of containers $(S,P) \to (T,Q)$ is a pair (u,v) with $u\colon S \to T$ a map of shapes and v a family $v_s\colon Q(u\,s) \to P(s)$ of position maps *going backwards*. This is exactly the data of a family, indexed by S, of objects of $\mathbf{Set}^{\mathrm{op}}$ (namely the $P(s)$) together with a map of the index sets and, over it, morphisms in $\mathbf{Set}^{\mathrm{op}}$. Hence

$$\mathbf{Cont} \;\cong\; \mathbf{Fam}(\mathbf{Set}^{\mathrm{op}}),$$

the *free coproduct completion* of $\mathbf{Set}^{\mathrm{op}}$: objects are S-indexed families of objects of $\mathbf{Set}^{\mathrm{op}}$, and $\mathbf{Fam}(\mathcal{A})$ is the universal category with all (small) coproducts containing $\mathcal{A}$. Under the extension functor $[\![-]\!]$, a family $(P(s))_{s\in S}$ maps to the coproduct of representables $\sum_{s\in S} y^{P(s)}$ (Remark 1.2).

Theorem 2.4 (Free coproduct completion as Lan; Neil's observation (i))**.** *Let $\mathcal{A}$ be a small category and let $Y\colon \mathcal{A} \to \mathbf{Fam}(\mathcal{A})$ be the "singleton" embedding sending a to the one-element family (a). Then $\mathbf{Fam}(\mathcal{A})$, with Y, is the free cocompletion of $\mathcal{A}$ under coproducts, and consequently:*

(a) *For any functor $F\colon \mathcal{A} \to \mathcal{E}$ into a category $\mathcal{E}$ with small coproducts, the* coproduct-preserving *extension $\widehat{F}\colon \mathbf{Fam}(\mathcal{A}) \to \mathcal{E}$ defined on objects by*

$$\widehat{F}\big((a_s)_{s\in S}\big) \;=\; \coprod_{s\in S} F(a_s)$$

is the left Kan extension $\mathrm{Lan}_Y F$*, and $\widehat{F} \circ Y \cong F$.*

(b) *In particular $\widehat{F}$* preserves representables*: it sends the singleton family $(a) = Y(a)$ to $F(a)$, i.e. to the generator $F(a)$, with no further identifications.*

Specialising $\mathcal{A} = \mathbf{Set}^{\mathrm{op}}$, $F = y^{(-)}\colon \mathbf{Set}^{\mathrm{op}} \to [\mathbf{Set},\mathbf{Set}]$ (the representable embedding), and $\mathcal{E} = [\mathbf{Set},\mathbf{Set}]$, the extension $\widehat{y^{(-)}}\colon \mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\mathrm{op}}) \to [\mathbf{Set},\mathbf{Set}]$ is exactly the container-extension functor $[\![-]\!]$, and it is the left Kan extension of the representable embedding along the singleton embedding Y.[1] *Thus*

$$[\![-]\!] \;=\; \mathrm{Lan}_Y\, y^{(-)}, \qquad [\![S,P]\!] \;=\; \coprod_{s\in S} y^{P(s)}.$$

Proof. We prove (a) and (b); the specialisation is then read off.

The coend collapses to the coproduct. Apply Theorem 2.2 with $K = Y\colon \mathcal{A} \to \mathbf{Fam}(\mathcal{A})$. For an object $d = (a_s)_{s\in S}$ of $\mathbf{Fam}(\mathcal{A})$,

$$(\mathrm{Lan}_Y F)(d) \;\cong\; \int^{a\in\mathcal{A}} \mathbf{Fam}(\mathcal{A})\big(Ya,\, d\big) \cdot F(a).$$

[1] A size remark, since $\mathbf{Set}^{\mathrm{op}}$ is *not* small and Theorem 2.2 was stated for small $\mathcal{C}$. Two things rescue the specialisation, and neither needs a large coend to converge. First, the objects of $\mathbf{Fam}(\mathbf{Set}^{\mathrm{op}}) = \mathbf{Cont}$ are indexed by *small* shape sets S, so the coproduct $\coprod_{s\in S} F(a_s)$ that the proof lands on is a small colimit, which exists in $[\mathbf{Set},\mathbf{Set}]$. Second — and this is the clean route — part (a) is really a consequence of the *universal property* of the free coproduct completion (coproduct-preserving functors out of $\mathbf{Fam}(\mathcal{A})$ biject with functors out of $\mathcal{A}$), which holds for large $\mathcal{A}$ verbatim; the coend computation above is the concrete face of that universal property, not a hypothesis we depend on.

By the definition of morphisms in the free coproduct completion, a map from a *singleton* family $Ya = (a)$ into $d = (a_s)_{s\in S}$ is a choice of one index $s \in S$ together with a morphism $a \to a_s$ in $\mathcal{A}$:

$$\mathbf{Fam}(\mathcal{A})\big(Ya,\, (a_s)_s\big) \;\cong\; \coprod_{s\in S} \mathcal{A}(a,\, a_s).$$

(There is no coproduct *to* match on the domain side — the domain is a singleton — which is exactly why singletons are the generators.) Substituting,

$$(\mathrm{Lan}_Y F)(d) \;\cong\; \int^a \Big(\coprod_{s\in S} \mathcal{A}(a, a_s)\Big) \cdot F(a) \;\cong\; \coprod_{s\in S} \int^a \mathcal{A}(a, a_s) \cdot F(a),$$

using that copowers and the coend (both colimits) commute with the coproduct over s. The inner coend is the co-Yoneda / density formula (Theorem 1.6) evaluated for the functor F at the object a_s:

$$\int^a \mathcal{A}(a, a_s) \cdot F(a) \;\cong\; F(a_s).$$

Hence $(\mathrm{Lan}_Y F)(d) \cong \coprod_{s\in S} F(a_s) = \widehat{F}(d)$, naturally in d. This proves (a): the coproduct-preserving extension *is* the left Kan extension, with unit $\eta\colon F \cong \widehat{F} \circ Y$ the identification of $F(a)$ with the one-term coproduct.

Preservation of representables (b). By (a), $\widehat{F}(Ya) = \coprod_{s\in 1} F(a) = F(a)$, on the nose. When $F = y^{(-)}$ the generators $F(a)$ are the representables themselves, so $\widehat{y^{(-)}}$ sends singletons to representables: it preserves representables.

Identification with $[\![-]\!]$. For $\mathcal{A} = \mathbf{Set}^{\mathrm{op}}$ and $d = (P(s))_{s\in S}$ a container (S, P), part (a) gives $\widehat{y^{(-)}}(S, P) = \coprod_{s\in S} y^{P(s)}$, whose value at X is $\sum_{s\in S} \mathbf{Set}(P(s), X) = [\![S, P]\!](X)$ by Remark 1.2. So $\widehat{y^{(-)}} = [\![-]\!] = \mathrm{Lan}_Y y^{(-)}$. $\square$

Remark 2.5 (Why this is exactly the reusable form the book needs)**.** Theorem 2.4 says the container-extension functor is not an ad hoc gadget: it is *forced* as the unique cocontinuous (coproduct-preserving) extension of the representable embedding. Two consequences the monoidal chapter uses without reproof: (i) any structure on $\mathbf{Set}^{\mathrm{op}}$ that we wish to transport to **Cont** need only be defined on generators (singletons) and extended by coproducts, and the result is automatically the Kan extension; (ii) to check that a functor out of **Cont** agrees with $[\![-]\!]$ it suffices to check agreement on representables plus preservation of coproducts. These are precisely the two hooks Theorem C hangs on.

Example 2.6 (The extension of a two-shape container)**.** Take $S = \{s_0, s_1\}$, $P(s_0) = 1$, $P(s_1) = 2 = \{0, 1\}$. Then

$$[\![S, P]\!] \;=\; y^1 \;\amalg\; y^2 \;\cong\; \mathrm{Id} \;\amalg\; (-)^2, \qquad [\![S, P]\!](X) \;=\; X \;+\; X^2.$$

This is the functor whose algebras are "a nullary-of-arity-1 and a binary operation" — e.g. the signature functor for magmas-with-a-unary-op. The point of Theorem 2.4 is that we did *not* compute a coend to get this: the completion being free, the Kan extension is literally the coproduct of representables indexed by shapes, read straight off (S, P).

3 Day convolution

Day convolution transports a monoidal structure from a category to its presheaf category, in the universal way compatible with Yoneda. On **Cont** it degenerates to "multiply shapes, combine positions," with no coend to compute — and that degeneration is the reason the next chapter's monoidal structures are as tractable as they are.

Definition 3.1 (Day convolution)**.** Let $(\mathcal{C}, \otimes, I)$ be a small monoidal category. The *Day convolution* of copresheaves $F, G\colon \mathcal{C} \to \mathbf{Set}$ is

$$(F \circledast G)(c) \;=\; \int^{c_1,c_2\in\mathcal{C}} \mathcal{C}(c_1 \otimes c_2,\, c) \times F(c_1) \times G(c_2),$$

with unit the representable $y^I = \mathcal{C}(I, -)$.

Theorem 3.2 (Day [1]; universal property)**.** *$([\mathcal{C}, \mathbf{Set}], \circledast, y^I)$ is a symmetric*[2] *monoidal category. It is cocontinuous in each variable, and the Yoneda embedding*[3]

$$y^{(-)}\colon (\mathcal{C}^{\mathrm{op}}, \otimes, I) \longrightarrow ([\mathcal{C}, \mathbf{Set}], \circledast, y^I)$$

is strong monoidal*: there are natural isomorphisms*

$$y^{c_1} \circledast y^{c_2} \;\cong\; y^{\,c_1\otimes c_2}, \qquad y^I = y^I.$$

Moreover Day convolution is the unique *monoidal structure on* $[\mathcal{C}, \mathbf{Set}]$ *that is cocontinuous in each variable and makes $y^{(-)}$ strong monoidal.*

Reference and the one computation we will reuse. The full statement is Day [1]; see also Loregian [2, Ch. 3]. We record only the representable computation, as the degeneration below leans on it. By the coend formula and the co-Yoneda lemma (density, Theorem 1.6) applied twice,

$$(y^a \circledast y^b)(c) = \int^{c_1,c_2} \mathcal{C}(c_1 \otimes c_2, c) \times \mathcal{C}(a, c_1) \times \mathcal{C}(b, c_2) \cong \mathcal{C}(a \otimes b,\, c) = y^{a\otimes b}(c),$$

the first coend variable collapsing $c_1 \rightsquigarrow a$ and the second $c_2 \rightsquigarrow b$ by density. Cocontinuity in each variable is immediate: coends and products-with-a-fixed-set are colimits and preserve colimits in F (resp. G). Uniqueness: any cocontinuous-in-each-variable monoidal structure is determined by its values on the dense generators (the representables) by Remark 1.7, and strong-monoidality pins those values to $y^a \circledast y^b \cong y^{a\otimes b}$; two such structures therefore agree everywhere. □

3.1 Key degeneration: Day on Cont needs no coend

Now we cash Theorem 2.4 against Theorem 3.2. The category carrying our presheaves is $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\mathrm{op}})$, the *free* coproduct completion of $\mathbf{Set}^{\mathrm{op}}$; its objects are coproducts of representables; Day is cocontinuous in each variable. So Day is determined on representables and extended by coproducts — and on representables it is just $\otimes$ of the representing objects. No coend survives.

[2] Symmetric when $\otimes$ is; in general braided/plain monoidal as $\otimes$ is.

[3] The domain is $\mathcal{C}^{\mathrm{op}}$, not $\mathcal{C}$: recall (§1) that $y^{(-)}\colon c \mapsto \mathcal{C}(c, -)$ is *covariant out of* $\mathcal{C}^{\mathrm{op}}$, a morphism $a \to b$ of $\mathcal{C}^{\mathrm{op}}$ inducing $y^a \Rightarrow y^b$ by precomposition. The tensor is unchanged on objects, so strong monoidality reads $y^a \circledast y^b \cong y^{a\otimes b}$ either way; we keep the variance honest because the container application turns exactly on it.

Proposition 3.3 (Degenerate Day on containers)**.** *Let* $(\mathbf{Set}, \star, I)$ *be any monoidal structure on* $\mathbf{Set}$ *(below:* $\star \in \{+, \times, \vee\}$*). Taking it as the base monoidal category* $\mathcal{C} = (\mathbf{Set}, \star)$ *of Definition 3.1 — the opposite enters only through the* embedding $y^{(-)} \colon \mathbf{Set}^{\mathrm{op}} \to [\mathbf{Set}, \mathbf{Set}]$ *and the identification* $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\mathrm{op}})$*, not through* $\star$ *— Day convolution induces a monoidal structure* $\odot_\star$ *on* $\mathbf{Cont}$ *given* without any coend *by*

$$p \ \odot_\star \ q \ = \ \Big(S_p \times S_q, \quad (s,t) \mapsto p[s] \star q[t] \Big),$$

where $p = (S_p, p[-])$ *and* $q = (S_q, q[-])$ *are containers (writing* $p[s]$ *for the position set at shape* s*). On extensions this is the Day convolution induced by* $\star$*; on representables it restricts to* $y^a \odot_\star y^b = y^{a \star b}$*.*

Proof. Write every container as a coproduct of representables, $p = \coprod_{s \in S_p} y^{p[s]}$, $q = \coprod_{t \in S_q} y^{q[t]}$ (Theorem 2.4). Day convolution is cocontinuous in each variable (Theorem 3.2), so it commutes with these coproducts:

$$p \circledast_\star q \ \cong \ \Big(\coprod_s y^{p[s]} \Big) \circledast_\star \Big(\coprod_t y^{q[t]} \Big) \ \cong \ \coprod_{s,t} \big(y^{p[s]} \circledast_\star y^{q[t]} \big).$$

By strong-monoidality on representables (the representable computation in the proof of Theorem 3.2, with base $\mathcal{C} = (\mathbf{Set}, \star)$ and the $\mathbf{Set}^{\mathrm{op}}$-valued embedding $y^{(-)}$), $y^{p[s]} \circledast_\star y^{q[t]} \cong y^{\,p[s] \star q[t]}$. Hence

$$p \circledast_\star q \ \cong \ \coprod_{(s,t) \in S_p \times S_q} y^{\,p[s] \star q[t]},$$

which is exactly the container $\big(S_p \times S_q,\, (s,t) \mapsto p[s] \star q[t]\big)$ read off by Theorem 2.4. The coend that defined $\circledast_\star$ has been discharged entirely by density on the generators plus cocontinuity; nothing integral remains. □

Remark 3.4 (Where the coend went)**.** The coend in Definition 3.1 is a colimit that identifies ways of factoring an object as $c_1 \otimes c_2$. On a *free* coproduct completion the only objects that need testing are the generators (singletons), and a generator factors through $\otimes$ in exactly one way up to the coend's identifications — that is the content of the representable collapse $y^a \circledast y^b \cong y^{a \otimes b}$. So the coend does no work beyond the generators, and cocontinuity carries the answer to all coproducts. This is the same phenomenon as Theorem 2.4: "free completion ⇒ structure is what it does on generators."

Example 3.5 (Three instances side by side)**.** Take $\star \in \{+, \times, \vee\}$ where $\vee$ is a fixed monoidal structure on $\mathbf{Set}$ (for concreteness one may read $\vee$ as a chosen "max-of-arities" combination; we do *not* classify these here — that is the business of the next chapter). Proposition 3.3 gives, for containers $p = (S_p, p[-])$, $q = (S_q, q[-])$:

$\star$	$p \ \odot_\star \ q$ (positions at shape (s,t))	one-line reading
$+$	$p[s] + q[t]$	a position on the left *or* on the right
$\times$	$p[s] \times q[t]$	a position on the left *and* on the right
$\vee$	$p[s] \vee q[t]$	the chosen combined arity

In all three cases shapes multiply, $S_p \times S_q$; only the position-combining law changes. Concretely with $p = y^2$ (one shape, two positions) and $q = y^3$: $p \odot_+ q = y^5$, $p \odot_\times q = y^6$, and $p \odot_\vee q = y^{\,2 \vee 3}$.

We stress that we have proved *the formula*, not any classification of which monoidal structures on **Cont** arise this way; the classification (Theorem C and its companions) is the next chapter's work, and the present degeneration lemma is the tool it stands on.

4 Yoneda, monoidal, and closed

The last piece is closed structure. A closed monoidal category has internal homs $[-,-]$ right adjoint to the tensor. Neil's observation (ii) is that closed structure on a Day category is *forced on representables and extended by density*: because representables are a dense generator, the internal hom is pinned down by its behaviour against y^c and has no freedom elsewhere. We prove exactly that, and we are deliberately careful about a temptation it invites and does not license. It is tempting to conclude that *any* cocontinuous strong-monoidal functor out of $[\mathcal{C}, \mathbf{Set}]$ automatically preserves the internal hom — "it agrees on generators, so it agrees everywhere." That inference is *false*: the internal hom is built from a *limit* (right adjoint), and a cocontinuous functor is under no obligation to preserve limits. We give a one-line counterexample (Remark 4.3). Preservation is a genuine, additional fact; it holds for the container-extension functor, and the monoidal chapter establishes *that* directly, using this section only for the density principle and the representable formula.

Definition 4.1 (Closed monoidal category)**.** A monoidal category $(\mathcal{V}, \otimes, I)$ is *closed* if for every object b the functor $-\otimes b$ has a right adjoint $[b,-]$, the *internal hom*, i.e. there are natural isomorphisms $\mathcal{V}(a \otimes b,\ c) \cong \mathcal{V}(a,\ [b, c])$.

Theorem 4.2 (Closure is determined on representables; Neil's observation (ii))**.** *Let $(\mathcal{C}, \otimes, I)$ be a small monoidal category and equip $[\mathcal{C}, \mathbf{Set}]$ with Day convolution $(\circledast, y^I)$ (Theorem 3.2).*

(a) (Day is closed.) $[\mathcal{C}, \mathbf{Set}]$ *is closed for $\circledast$: each $-\circledast G$ has a right adjoint $[G,-]_\circledast$, and*

$$[G, H]_\circledast(c) \ \cong\ \mathrm{Nat}\big(y^c \circledast G,\ H\big).$$

The internal hom is thus determined by its behaviour on representable first arguments*; on a representable it is a shift, $[y^c, H]_\circledast \ \cong\ H(c \otimes -)$, and it is extended to all G by density (Theorem 1.6).*

(b) (The hom of two representables.) *Evaluated on a pair of representables,*

$$[\,y^a,\ y^b\,]_\circledast(d) \ \cong\ \mathrm{Nat}\big(y^{d\otimes a},\ y^b\big) \ \cong\ \mathcal{C}(b,\ d \otimes a).$$

This is in general a coproduct *of representables — a genuine container —* not *a single representable. For the Dirichlet tensor $\star = \times$ on* **Set** *(Proposition 3.3) it is the container whose shapes are the maps $b \to a$, i.e. the morphisms* $\mathbf{Cont}(y^a, y^b)$*, with b positions apiece; that is precisely the "shapes are morphisms" internal hom of the monoidal chapter.*

Part (a) is Day's [1]. What part (a) does not *assert is that an arbitrary functor sharing these values preserves the internal hom; see Remark 4.3 and Corollary 4.4 for the honest form of the transport the book uses.*

Proof. (a) The Day-closed structure. Fix G. The functor $-\circledast G$ is cocontinuous in its first variable (Theorem 3.2), and $[\mathcal{C}, \mathbf{Set}]$ is a presheaf category, hence cocomplete and locally presentable; a cocontinuous endofunctor of a locally presentable category has a right adjoint (adjoint functor theorem for locally presentable categories, Riehl [4, §E]). Call it $[G,-]_\circledast$. To identify it on representables, use Yoneda and the adjunction:

$$[G,H]_\circledast(c) \;\cong\; \mathrm{Nat}\big(y^c,\, [G,H]_\circledast\big) \;\cong\; \mathrm{Nat}\big(y^c \circledast G,\; H\big),$$

the first step by the Yoneda lemma (Theorem 1.3) and the second by the $(-\circledast G) \dashv [G,-]_\circledast$ adjunction. This proves the displayed formula and shows $[G,-]_\circledast$ is *determined* by its behaviour against representables — exactly "determined on representables and extended by density." Taking $G = y^c$ representable and reading the same formula in the first slot,

$$[y^c, H]_\circledast(d) \cong \mathrm{Nat}\big(y^d \circledast y^c,\, H\big) \cong \mathrm{Nat}\big(y^{d\otimes c},\, H\big) \cong H(d \otimes c),$$

by Yoneda and $y^d \circledast y^c \cong y^{d\otimes c}$ (Theorem 3.2); so $[y^c, H]_\circledast \cong H((-)\otimes c)$ is a shift, as claimed.

(b) The hom of two representables. Put $H = y^b$ in the shift formula from part (a):

$$[y^a, y^b]_\circledast(d) \cong y^b(d \otimes a) = \mathcal{C}(b,\, d\otimes a),$$

using $[y^a, y^b]_\circledast(d) \cong \mathrm{Nat}(y^d \circledast y^a, y^b) \cong \mathrm{Nat}(y^{d\otimes a}, y^b) \cong \mathcal{C}(b, d\otimes a)$ (Yoneda and strong monoidality). Whether this is again representable is exactly the question of whether $d \mapsto \mathcal{C}(b, d\otimes a)$ is a single representable, which it need not be; it is, however, always a coproduct of representables by density (Theorem 1.6), hence a container. For $\mathcal{C} = (\mathbf{Set}, \times)$,

$$[y^a, y^b]_\circledast(d) = \mathbf{Set}(b,\, d\times a) = \mathbf{Set}(b,d) \times \mathbf{Set}(b,a) = \coprod_{\phi\in\mathbf{Set}(b,a)} \mathbf{Set}(b,d) = \coprod_{\mathbf{Cont}(y^a,y^b)} y^b(d),$$

using $\mathbf{Cont}(y^a, y^b) \cong \mathbf{Set}(b,a)$ (Corollary 1.4). So the internal hom is the container with shape set $\mathbf{Cont}(y^a, y^b)$ and b positions at each shape — the "shapes are morphisms" formula the monoidal chapter certifies (after Niu–Spivak, Exercise 4.78). It is emphatically not y^{b^a}. $\square$

Remark 4.3 (Density determines; it does *not* transport for free)**.** It is tempting to upgrade Theorem 4.2 to: *any* cocontinuous strong-monoidal $L\colon [\mathcal{C}, \mathbf{Set}] \to \mathcal{V}$ (with $\mathcal{V}$ closed) preserves the internal hom, since it agrees with the target's internal hom on the dense generators. **This is false**, and the reason is worth internalising, because it recurs. The would-be argument reduces G to a coproduct of representables $G = \coprod_s y^{c_s}$ and uses that internal-hom-out-of-G is a *product* $\prod_s [y^{c_s}, -]$; but a cocontinuous L preserves *co*products, and is under no obligation to preserve that product. Internal hom is a right adjoint — a limit-shaped thing — and colimit-preservation buys nothing against it.

A one-line counterexample. Take $\mathcal{C} = \mathbf{1}$, so $[\mathcal{C}, \mathbf{Set}] = \mathbf{Set}$ with Day tensor the cartesian product and internal hom $[G,H] = H^G$. Let $\mathcal{V} = \mathrm{Vect}_k$ (closed under $\otimes_k$) and $L = k[-]$ the free-vector-space functor: it is cocontinuous (a left adjoint) and strong monoidal, $k[A\times B] \cong k[A] \otimes k[B]$, $k[1] = k$. The comparison is

$$\lambda_{G,H}\colon\; k[H^G] \;\longrightarrow\; \mathrm{Hom}_k\big(k[G], k[H]\big), \qquad (f\colon G \to H) \;\longmapsto\; \big(e_g \mapsto e_{f(g)}\big).$$

For G infinite and $|H| \geq 2$ (say $G = \mathbb{N}$, $H = \{0,1\}$) it is *not* surjective: a general linear map $k[\mathbb{N}] \to k[H]$ assigns arbitrary, independent images to the countably many basis vectors, whereas a *finite* linear combination of the "function" maps $e_g \mapsto e_{f(g)}$ realises only finitely many distinct column-patterns. So λ is not invertible: $L = k[-]$ is strong monoidal but not strong closed. What distinguishes it from the container-extension functor is that $k[-]$ is not fully faithful, whereas $[\![-]\!]$ is; and it is the honest, checkable preservation — not an abstract entailment — that the next corollary rests on.

Corollary 4.4 (The container reading)**.** *For the Dirichlet tensor $\odot_\times$ on* **Cont** $\cong$ **Fam**$(\mathbf{Set}^{\mathrm{op}})$*, Theorem 4.2 pins the internal hom down with no coend: it is determined on representables by the shift $[y^a, H]_\circledast \cong H((-)\times a)$, and on two representables it is the container*

$$[y^a, y^b]_\circledast \;=\; \Big(\mathbf{Cont}(y^a, y^b),\; b \text{ positions at each shape}\Big),$$

extended to all containers by density. That $\odot_\times$ is genuinely closed — that this candidate internal hom really is right adjoint to $-\odot_\times q$ — is the content the monoidal chapter certifies directly (after Niu–Spivak, Exercise 4.78; machine-checked in `DirichletClosed.lean`). Chapter 0 supplies only the two pieces that make the formula forced*: the density principle and the representable computation above. It does* not *supply closure "for free" from cocontinuity — Remark 4.3 shows why no such free lunch exists.* (This is the honest form the monoidal chapter cites.)

Example 4.5 (The Dirichlet hom of two representables, computed)**.** Take $\star = \times$. A natural first guess for $[y^a, y^b]_\circledast$ is the "pointwise exponential" y^{b^a} — and it is *wrong*, instructively so. Part (b) computes the honest answer:

$$[y^a, y^b]_\circledast(d) = \mathbf{Set}(b, d\times a) = \underbrace{\mathbf{Set}(b,d)}_{=\,y^b(d)} \times \mathbf{Set}(b,a) = \coprod_{\phi\colon\, b\to a} y^b(d).$$

So the internal hom is $\coprod_{\mathbf{Set}(b,a)} y^b$: a container with $|\mathbf{Set}(b,a)|$ shapes — one per morphism $y^a \to y^b$ — each carrying b positions. Take $a = 2$, $b = 1$, d arbitrary: the guess $y^{b^a} = y^{1^2} = y^1 = \mathrm{Id}$ gives $[\,\cdot\,](d) = d$, whereas the truth $\mathbf{Set}(1, d\times 2) = d\times 2$ gives $2d$; already off by a factor of two. The exponent lives in the *shapes* (the morphisms), not in the positions — which is exactly why the monoidal chapter's internal hom has "shapes are morphisms." We flag — for the next chapter — that at the level of the full functor category $[\mathbf{Set}, \mathbf{Set}]$ this Dirichlet-closed structure is *not* the container-composition structure $\triangleleft$; the present chapter asserts only that, whatever Day tensor we transported, its internal hom is *forced* on representables and extended by density.

What this chapter bought us

Three reusable facts, each proved from the same two ideas (Yoneda + density):

- **Representables generate.** Every functor is a canonical colimit of representables (Theorem 1.6); representables are a dense generator of $[\mathbf{Set}, \mathbf{Set}]$.
- **Completion is Kan.** The container-extension functor is the left Kan extension of the representable embedding along the singleton inclusion, i.e. free coproduct completion, and it preserves representables (Theorem 2.4). Hence Day convolution on **Cont** needs no coend (Proposition 3.3): $p \odot_\star q = (S_p \times S_q,\ (s,t) \mapsto p[s] \star q[t])$.
- **Closure is determined, not free.** A Day tensor is closed, and its internal hom is *forced* on representables — $[y^a, H]_\circledast \cong H((-)\otimes a)$, and $[y^a, y^b]_\circledast$ the "shapes are morphisms" container — then extended by density (Theorem 4.2). Density *determines* the hom; it does not hand it over to every cocontinuous functor for free (Remark 4.3), and the actual closure of the container tensors is certified directly in the monoidal chapter.

These are the pre-container tools the monoidal chapter's Theorem C and its companions stand on. Nothing here is new; the arrangement — and the care about what density does and does not buy — is ours.

References

[1] B. Day, *On closed categories of functors*, in Reports of the Midwest Category Seminar IV, Lecture Notes in Mathematics 137, Springer, 1970, pp. 1–38.

[2] F. Loregian, *(Co)end Calculus*, London Mathematical Society Lecture Note Series 468, Cambridge University Press, 2021.

[3] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics 5, Springer, 1998 (esp. Ch. III on Yoneda and Ch. X on Kan extensions).

[4] E. Riehl, *Category Theory in Context*, Aurora: Dover Modern Math Originals, Dover, 2016.